minimize the total transportation cost. The supply limits at the origins and the demand requirements at the destinations must be satisfied.

The above transportation problem can be written in the following tabular form:

		Destinations					
		1	2	3	…	n	supply
Origins	1	(x_{11}) C_{11}	(x_{12}) C_{12}	(x_{13}) C_{13}	…	(x_{1n}) C_{1n}	a_1
	2	(x_{21}) C_{21}	(x_{22}) C_{22}	(x_{23}) C_{23}	…	(x_{2n}) C_{2n}	a_2
	⋮		⋮	⋮	⋮	⋮	⋮
	m	(x_{m1}) C_{m1}	(x_{m2}) C_{m2}	(x_{m3}) C_{m3}	…	(x_{mn}) C_{mn}	a_m
demand		b_1	b_2	b_3	…	b_n	

Table-10.1

Now the linear programming model representing the transportation problem is given by

The objective function is Minimize $Z = \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}x_{ij}$ Subject to the constraints

$\sum_{j=1}^{n} x_{ij} = a_i$, i=1,2,…….m (supply constraints)

$\sum_{i=1}^{m} x_{ij} = b_j$, j=1,2,…….$n$ (demand constraints)

$x_{ij} \geq 0$ for all i,j. (non-negative restrictions)

Some Definitions

Feasible Solution: A feasible solution to a transportation problem is a set of non-negative values x_{ij}(i=1,2,..,m, j=1,2,…n) that satisfies the constraints.

Basic Feasible Solution: A feasible solution is called a basic feasible solution if it contains not more than $m+n-1$ allocations, where m is the number of rows and n is the number of columns in a transportation problem.

Optimal Solution: Optimal Solution is a feasible solution (not necessarily basic) which optimizes(minimize) the total transportation cost.

Non degenerate basic feasible Solution: If a basic feasible solution to a transportation problem contains exactly $m+n-1$ allocations in independent positions, it is called a Non degenerate basic feasible solution. Here m is the number of rows and n is the number of columns in a transportation problem.

Degeneracy : If a basic feasible solution to a transportation problem contains less than $m+n-1$ allocations, it is called a degenerate

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